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Electronic structures and Mott state of epitaxial TaS_2 monolayers

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Layered material TaS_2 hosts multiple structural phases and exotic correlated quantum states, including charge density wave (CDW), superconductivity, quantum spin liquid, and Mott insulating state. Here, we synthesized TaS_2 monolayers in H and T phases using the molecular beam epitaxial (MBE) method and studied their electronic structures via angle-resolved photoemission spectroscopy (ARPES). We found that the H phase TaS_2 (H-TaS_2) monolayer is metallic, with an energy band crossing the Fermi level. In contrast, the T phase TaS_2 (T-TaS_2) monolayer shows an insulated energy gap at the Fermi level, while the normal calculated band structure implies it should be metallic without any band gap. However, by considering Hubbard interaction potential U, further density functional theory (DFT) calculation suggests that monolayer T-TaS_2 could be a CDW Mott insulator, and the DFT+U calculation matches well with the ARPES result. More significantly, the temperature-dependent ARPES result indicates that the CDW Mott state in the T-TaS_2 monolayer is more robust than its bulk counterpart and can persist at room temperature. Our results reveal that the dimensional effect can enhance the CDW Mott state and provide valuable insights for further exploring the exotic properties of monolayer TaS_2.

TaS_2, Mott insulating state, charge density wave, angle-resolved photoemission spectroscopy

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1 Introduction

Since the discovery of graphene, layered materials have drawn tremendous attention due to their exotic properties in two-dimensional (2D) limit [1]. As a major branch of layered materials, layered transition metal dichalcogenides (TMDCs) exhibit plenty of exciting phenomena, ranging from charge density wave (CDW) and superconductivity to topological phase, providing a great platform to study the interplay between interlayer coupling, electron-electron and electron-phonon interactions [2,3]. Recent advances in mechanical exfoliation and thin film growth allow the exploration of samples as thin as one unit cell [1,4]. The lack of screening along the out-of-plane direction can also dramatically enhance the electron-electron interaction [5]. Thinning-induced dimensional crossover can strongly affect the electronic structure of a material, such as the indirect-to-direct band gap transition in $MoSe_2$ [6], metal-to-insulator transition in $PtTe_2$ [7], and 2D topological insulating phase in monolayer WTe_2 [8].

Among all TMDCs, the layered TaS_2 has attracted great research interest for a long time in both H and T phases

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[9-12]. For the H phase, bulk 2H-TaS_2 was reported to exhibit 3 × 3 CDW in layers below 75 K [13,14] and superconductivity transition at around 1 K [15]. The superconductivity transition temperature can be enhanced by either applying high pressure or thinning thickness [16,17]. For the T phase of TaS_2, multiple CDW states and transitions were reported. With cooling, bulk 1T-TaS_2 first enters an incommensurate CDW (I-CCDW) phase below 550 K, then a nearly commensurate CDW (NC-CDW) phase below 350 K, last a $\sqrt{13} \times \sqrt{13}$ R13.9° commensurate CDW (C-CDW) phase with an insulating ground state below 180 K [18,19]. The origin of the insulating state has been controversial, which is commonly associated with a Mott phase [20-25], while alternative explanations based on the three-dimensional stacking order of the CDW have been proposed recently [26-28]. Superconductivity can also be realized upon high pressure or doping [29-31]. Additionally, magnetic order and possible quantum spin liquid ground state were also predicted and explored [32,33]. Recent angle-resolved photoemission spectroscopic (ARPES) results of bulk 1T-TaS_2 show significant energy dispersions along the out-of-layer plane direction [34-36], which reflects the strong interlayer electronic coupling. Scanning tunneling microscopy/spectroscopy (STM/STS) results also show that the local electronic density of states (DOS) varies for different terraces [37-39]. An electronic structure study of monolayer TaS_2 where the interlayer coupling is absent would provide a deeper understanding of these exotic properties.

In this study, we epitaxially grew H and T phase TaS_2 monolayers on bilayer graphene (BLG) using molecular beam epitaxy (MBE) and investigated their band structures via ARPES. We found that the H phase TaS_2 (H-TaS_2) monolayer exhibited a metallic band structure with an energy band crossing the Fermi level, while the T phase TaS_2 (T-TaS_2) monolayer is an insulator with a bandgap opening at Fermi level. Complementary first-principles calculations based on density functional theory plus Hubbard interaction potential U (DFT + U) suggest that monolayer T-TaS_2 is indeed a Mott insulator in the CDW state, and the calculated band structure is in good agreement with the ARPES spectra. Furthermore, temperature-dependent ARPES results reveal that the grown T-TaS_2 monolayer can maintain its CDW Mott state from 9 K to room temperature, which differs from the behavior observed in the bulk counterpart (where the transition occurs at around ~200 K) [19,35]. These results provide a window into the electronic structure of monolayer TaS_2 and will also help to clarify the long-standing debate over the CDW and Mottness in 1T-TaS_2.

2 Methods

The growth of the TaS_2 monolayer was performed in a customed MBE ultra-high vacuum (UHV) system with a base pressure of ~2 × 10^{-10} mbar. The BLG substrates were prepared by repeatedly flash annealing 4H-SiC (0001) wafers to 1250°C for 60 cycles [40]. The TaS_2 samples were grown on the prepared BLG substrate by co-depositing Ta and S. The Ta flux was produced from an electron-beam evaporator. The S flux was supplied by the thermal decomposition of FeS using a standard Knudsen cell. The growth procedure was monitored by an *in-situ* reflection high energy electron diffraction (RHEED) system. After the growth, the samples were annealed at 500°C in sulfur background for 20 min. The sample surface was then characterized by an *in-situ* room-temperature (RT-) STM system. The APRES and X-ray photoemission spectroscopic (XPS) measurements were performed using a shared Scienta Omicron DA30L analyzer. The ultraviolet (UV) light source was generated by a Helium discharge lamp with a monochromator (He I mode, $h\nu$ = 21.218 eV). The overall energy resolution was ~20 meV. The monochromatic X-ray was generated from an Al electrode excitation source (Alα, 1486.7 eV). During the ARPES and XPS measurements, the samples were cooled down to the lowest ~9 K by a closed-cycle cryogenerator.

The DFT calculations were performed in the Vienna *ab initio* simulation package [41,42]. A large vacuum layer (~25 Å) was added to avoid image interactions due to periodic boundary conditions. The Perdew-Burke-Ernzerhof (PBE) generalized gradient approximation (GGA) + U [43,44] and the projector-augmented wave method [45] were used with plane-wave energy cut-off set to 450 eV. The CDW supercell was obtained by manually moving the position of the surrounding 12 Ta atoms to the center and then relaxing. The unfolding of the band structure from the CDW supercell to the undistorted unit cell was calculated using Vaspkit [46]. The Brillouin zone was sampled with the 6 × 6 × 1 Monkhorst Pack mesh for the primitive cell and a 3 × 3 × 1 mesh for the supercell.

3 Results and discussions

Figure 1(a) and (b) illustrate the crystal structure of single-layer T phase and H phase TaS_2, which consists of a layer of Ta atoms sandwiched by two layers of S atoms. The six S atoms adopt a trigonal prism configuration for H-TaS_2 while an octahedral one for the T-TaS_2. Graphene, known for its suitability as a substrate for van der Waals epitaxy, was utilized for the growth of TaS_2 films. The RHEED pattern of the BLG substrate is shown in Figure 1(c), and the RHEED pattern of a grown TaS_2 sample with sub-monolayer coverage is shown in Figure 1(d). The sharp (1 × 1) diffraction stripes of the grown sample indicate the good crystallization of TaS_2. The spacing between the (1 × 1) diffraction peaks in the RHEED pattern was measured to extract the in-plane

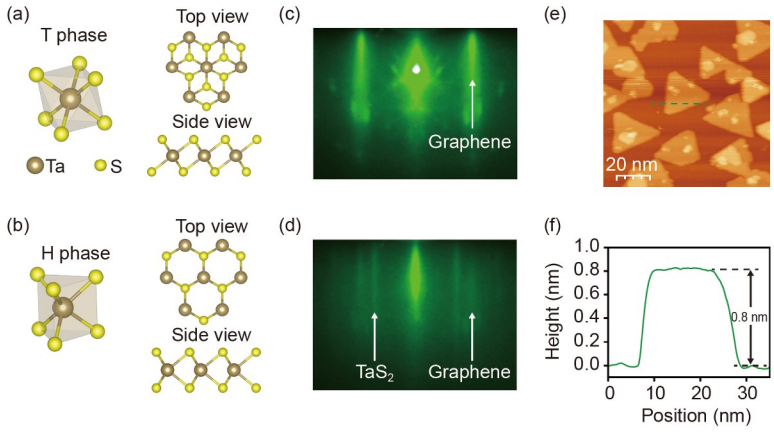

Figure 1 (Color online) (a), (b) Schematic atomic structures of T-TaS_2 and H-TaS_2 monolayers, respectively. (c) RHEED pattern of a clean BLG/SiC substrate. (d) RHEED pattern of a TaS_2 sample grown at 450°C with a submonolayer coverage. (e) STM image of the grown TaS_2 sample. (f) The line-scan profile was measured along the green dashed line in (e) across the TaS_2 island.

lattice constant, which was determined to be ~3.3 Å. The lattice constant of the grown TaS_2 monolayer is in good agreement with the values for the bulk counterpart [9], suggesting a weak interface strain effect and van der Waals-type coupling between the BLG substrate and grown film. Distinguishing the H and T phases from the RHEED pattern is challenging due to their almost the same lattice constants [9]. An STM topographic image of the sample grown at 450°C is shown in Figure 1(e), and more STM images of the TaS_2 samples grown at different temperatures can be found in the Supplementary Material, part A. The step height of the grown TaS_2 monolayer is ~0.8 nm (Figure 1(f)), which is typical for monolayer TMDC grown on BLG substrate [47]. According to the previous report, the H and T phases also share a close step height and can be distinguished by scanning tunneling spectroscopy (STS) or different CDW order [48]. There are some Ta clusters formed on the monolayer islands, which is speculated to occur due to the residence time of sulfur becomes much shorter at elevated substrate temperature [49]. To verify this point, we also performed the growth of TaS_2 on $SrTiO_3$(111) substrate, which can provide much stronger absorbability. The results are shown in Supplementary Material, part B. The TaS_2 sample grown on $SrTiO_3$(111) substrate is cluster-free, but the grain sizes are smaller. Besides, the TaS_2 grown on $SrTiO_3$(111) is all T phase.

The distinct band structure difference between the H and T phases would be a feasible way to determine the structural phases of the grown TaS_2 monolayer samples. According to a previous report, the ratio of H phase to T phase TaS_2 can be controlled via the substrate temperature and the overall coverage [48]. Here, we use ARPES to reveal the band structure difference of the TaS_2 samples grown at different substrate temperatures T_{Sub} = 25, 450, and 650°C. The measured ARPES spectra show two distinct features corresponding to different structural phases. Figure 2(a) shows the ARPES spectra of TaS_2 monolayer grown at T_{Sub} = 25°C. One can see an electron band crossing the Fermi level, as well as some other valence bands below −1.4 eV. A DFT calculation of monolayer H-TaS_2 is presented in Figure 2(b). The ARPES result in Figure 2(a) shows an overall agreement with the calculation result in Figure 2(b) despite the discrepancy of energy positions due to the weak description of electron correlations in the GGA exchange-correlation functional. The observed band structure in ARPES is also consistent with the features in a previous report on monolayer H-TaS_2 grown on Au(111), except for less electron doping from the substrate [50]. These findings suggest that the TaS_2 monolayer grown at a low substrate temperature of T_{Sub} = 25°C is the H phase. It is reported TaS_2 monolayer also exhibits a 3 × 3 CDW at 4.6 K and disappears at the liquid nitrogen temperature of 77 K [48]. The ARPES spectra taken at 9 and 100 K can be found in Supplementary Material, part C. Unfortunately, the CDW-induced change in band structure is too small to be distinguished.

The ARPES spectra of the TaS_2 monolayer grown at T_{Sub} = 450°C is shown in Figure 2(c). In addition to the bands of the H-TaS_2 monolayer, some additional bands emerge. A new hole band with a flat feature around the Γ point appears at −0.25 eV, and a highly dispersive hole band appears below −1.56 eV. These bands are likely to be of T phase in origin. When the substrate temperature is further increased to 650°C, the band intensity from the H phase nearly disappears and only the T-TaS_2 band can be observed (Figure 2(e)). These results suggest that T phase TaS_2 has a higher nucleation temperature compared with the H phase, which is in agreement with the previous study [48]. We found that the H and T phases of TaS_2 monolayer grow mixedly in a large temperature window (from 200°C to 600°C). The single-phase H-TaS_2 or T-TaS_2 can be obtained at near room temperature or above 650°C, respectively. The H and T phases of the TaS_2 monolayer can also be distinguished from their core

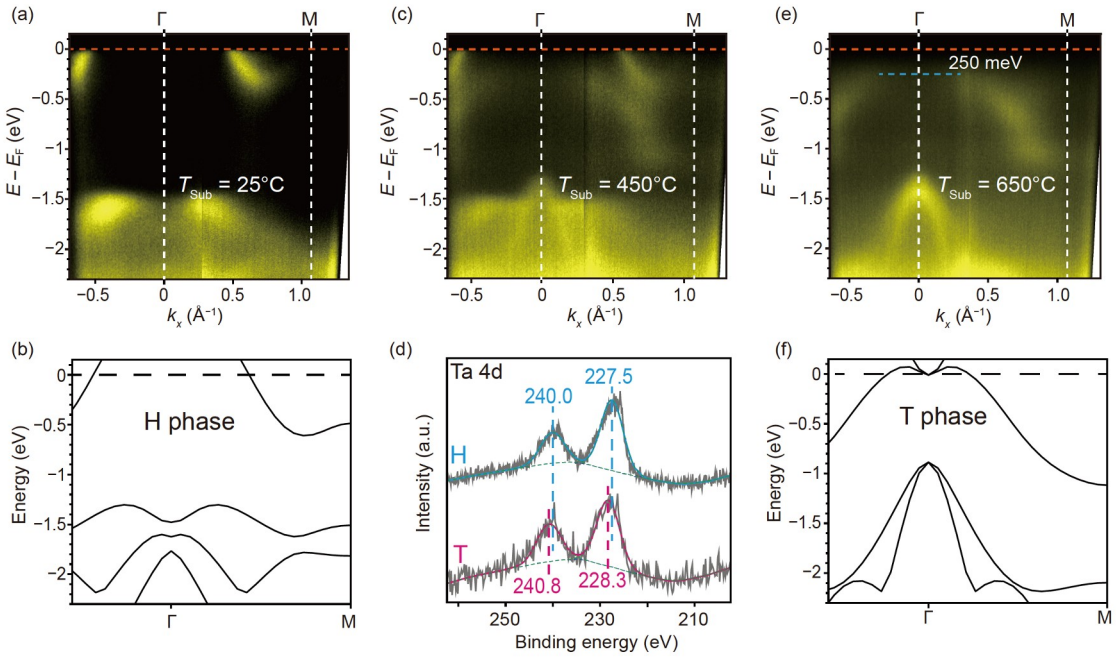

Figure 2 (Color online) (a) ARPES spectra of H-TaS_2 monolayer grown at T_{Sub} = 25°C. (b) Calculated band structure of monolayer H-TaS_2. (c) ARPES spectra of H/T mixed phase TaS_2 monolayer grown at T_{Sub} = 450°C. (d) XPS spectra around Ta 4d orbital of H and T phase TaS_2. (e) ARPES spectra of T-TaS_2 monolayer grown at T_{Sub} = 650°C. (f) Calculated band structure of monolayer T-TaS_2 without considering the CDW Mott phase.

level spectra in XPS. The Ta 4d peaks of T-TaS_2 exhibited a shift of ~0.8 eV towards higher binding energies as compared to the H-TaS_2 (Figure 2(d)). In Figure 2(e), the ARPES intensity near the Fermi level is strongly suppressed, indicating that the grown T-TaS_2 monolayer is an insulator. The calculated band structure of the T-TaS_2 monolayer without any CDW reconstruction (normal phase) is shown in Figure 2(f). The highly dispersive band at Γ point is mostly contributed by the S 3p orbit and is often referred to as S 3p band. The energy position of the S 3p band is also erroneously evaluated, probably due to GGA. Besides, the calculation results suggest that the 1T phase should be metallic with a band crossing the Fermi level, while the ARPES result in Figure 2(e) shows a small gap with the valence band maximum (VBM) at ~−250 meV below Fermi level (the method of determining VBM can be seen in Supplementary Material, part D). Simply moving the energy position of the Fermi level still cannot resolve this contradiction. This observed gap in T-TaS_2 is often attributed to a Mott gap [20-24], which will also be discussed below.

1T-TaS_2 hosts a unique $\sqrt{13} \times \sqrt{13}$ R13.9° CDW order. The 12 surrounding Ta atoms are pushed to the center 13th Ta atom, forming a "Star of David" cluster. A schematic of the CDW order and corresponding Brillouin zones of supercell and primitive cells are shown in Figure 3(a) and (b). The calculated band structure of the monolayer T-TaS_2 supercell in the CDW state is shown in Figure 3(c). According to the calculation, there is a band with a very narrow bandwidth centered at the Fermi level. When the ratio between the on-site Coulomb correlation energy U and the one-electron bandwidth W exceeds a critical value, a metal-to-Mott insulator transition would occur [51]. As shown in Figure 3(d), the GGA + U with spin polarization calculation result demonstrates that the narrow band at the Fermi level would split into two bands, and the system becomes an insulator when the on-site Coulomb energy U is considered. The gap size increases with increasing U and more details about the evolution of band structure with different U can be found in the Supplementary Material, part E. The calculation results of a supercell require an unfolding procedure to be compared with the APRES results [52]. Figure 3(e) and (f) show the comparison between the unfolded calculation results with U = 3 eV and the second derivative ARPES spectra of the T-TaS_2 monolayer. The unfolded DFT + U band structure shown in Figure 3(f) with U = 3 eV captures the major ARPES features, including the flat band near Γ point and the band discontinuity along the Γ-M direction, although the energy position of the S 3p band is still slightly misvalued. The band structure of the non-CDW T-TaS_2 with increasing U is shown in Supplementary Material, part F. Because of the large bandwidth near the Fermi level, adding a U = 3 eV will not cause a gap opening without the CDW configuration. The CDW-induced electron localization is important for the Mott transition to occur. In addition, the GGA + U calculation results also suggest the existence of local spin moments. However, according to the calculation results, the magnetic moment will be only 0.077 μB per Ta atom (1 μB per Star of David), which is very weak and might be very hard to detect

experimentally. Further surface-sensitive techniques, such as spin-resolved STM, may help to verify this point.

Previous ARPES works mentioned the ground state of bulk 1T-TaS_2 at low temperatures might be a band insulator due to the interlayer dimerization and the Mott insulating state that only exists in a small temperature window up on heating [35,53]. To see the temperature evolution of band structure in the 2D limit, we carried out temperature-dependent ARPES measurements of the grown T-TaS_2 monolayer sample. Figure 4(a)-(d) show the ARPES spectra of T-TaS_2 monolayer taken at 9, 100, 200, and 300 K, respectively. Unlike the results obtained from the bulk counterpart, the varying temperature does not induce a prominent change to the band structure. This can also be seen in the energy distribution curves (EDCs) extracted at the Γ point from the ARPES spectra (Figure 4(e)). The energy position of the VBM did not show an obvious shift with the variation of sample temperature. At different given temperatures, the VBM is always located at −250 meV below the Fermi level. The only noticeable change is the thermal broadening effect with increasing temperature. It's not surprising because the band gap in the T-TaS_2 monolayer is primarily determined by the on-site Coulomb energy U, and temperature should have a minor effect on it. The ratio between on-site Coulomb interaction U and bandwidth W is crucial for the Mott transition. When the dimensionality is reduced, the Coulomb energy U can be enhanced by reduced dielectric screening effect, and W is also suppressed by losing the dispersion

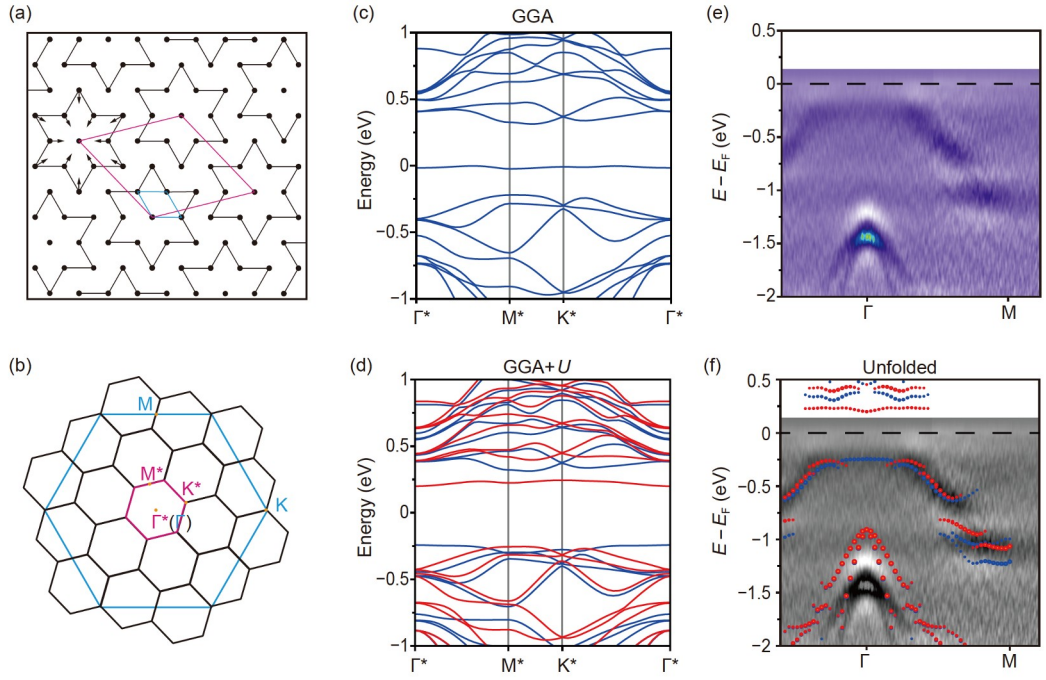

Figure 3 (Color online) (a) Unit cell of the $\sqrt{13} \times \sqrt{13}$ reconstruction of T-TaS_2. (b) Brillouin zone for T-TaS_2 and the $\sqrt{13} \times \sqrt{13}$ supercell. (c) GGA band structure of C-CDW phase of monolayer T-TaS_2. (d) GGA + U band structure of C-CDW phase of monolayer T-TaS_2 with U = 3 eV. (e) Second derivative ARPES spectra of the grown T-TaS_2 monolayer. (f) The calculated unfolded band structure with U = 3 eV of monolayer T-TaS_2 superimposed with the spectra in (e). The blue and red colors indicate the spin-up and spin-down states, and the polarization direction is along the out-of-plane direction.

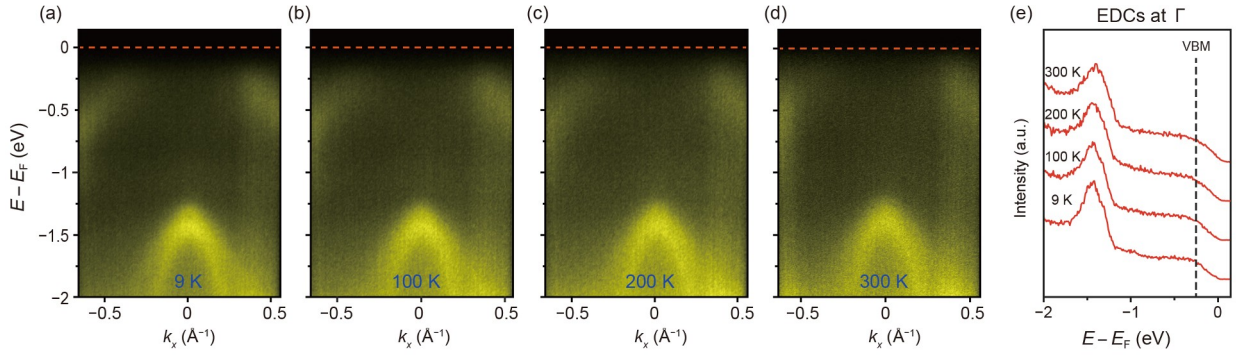

Figure 4 (Color online) ARPES spectra of the grown T-TaS_2 monolayer taken at (a) 9, (b) 100, (c) 200, and (d) 300 K, respectively. (e) EDCs at the Γ point extracted from the ARPES spectra.

along the out-of-plane direction [54-56]. For the multilayer T-TaS_2, the gap opening is a complex issue that is related to both strong correlation and interlayer interaction. We also present the calculation results of bilayer T-TaS_2 in Supplementary Material, part G. It is also interesting to refer to similar CDW materials such as monolayer 1T-$TaSe_2$ and 1T-$NbSe_2$ [54-57]. The main difference between TaS_2 and $TaSe_2$ is that the chalcogen dominant band hybridizes with the Ta dominant band only in the Se case. The contribution of Se electrons may affect the transport properties or screening effect [58]. Previous studies involving transport [59,60], STM [48,61], and transmission electron microscopy (TEM) [62] also revealed that the resistance-temperature transition points and stage of NC-CDW disappear in ultrathin samples, which is consistent with our ARPES results. In the 2D limit, the T-TaS_2 directly enters the C-CDW phase from the IC-CDW phase with cooling. Our finding demonstrates that although the bulk 1T-TaS_2 might be a band insulator at low temperature and undergoes multistage IC-NC-C CDW phase transitions related to interlayer interaction, and the monolayer T-TaS_2 is a rather simple system that is still a possible candidate for Mott physics and even the quantum spin liquid state.

4 Conclusion

To summarize, we have investigated the electronic structures of the MBE-grown H and T phase TaS_2 monolayers using ARPES. The ARPES results show that the H-TaS_2 monolayer is metallic, with an electron band crossing the Fermi level. In contrast, the T-TaS_2 monolayer holds an insulating state due to the CDW Mott state. The DFT + U calculation reproduces the CDW Mott band structure of monolayer T-TaS_2 well. The temperature-dependent ARPES results suggest that the complex multiple phase transitions in bulk TaS_2 have a strong relation with interlayer hopping, while monolayer 1T-TaS_2 persists in the CDW Mott state to at least room temperature. Our findings will also benefit the future applications of TMD materials in next-generation electronic devices.

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Conflict of interest The authors declare that they have no conflict of interest.

Supporting Information

The supporting information is available online at http://phys.scichina.com and https://link.springer.com. The supporting materials are published as submitted, without typesetting or editing. The responsibility for scientific accuracy and content remains entirely with the authors.

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